Preparing data for analysis Activity Book

Create Visualizations with Qlik Sense

CVQS May 2023

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Activity 1: Prepare data for a new app

Summary:

- Create a new Qlik Sense app and add app data from Excel.
- Use Data manager to prepare the tables for analysis with source filters, field transformations, and new field creations.
- Gain familiarity with the Visual preparation (Data manager associations) to associate the tables and create the data model.

Qlik Sense app: None, user creates a new app, ABC Sales.

Data Files: Qlik Sense Data\ExcelFiles**SalesData_MultiTabs.xlsx**

Challenge

Create a new app called **ABC Sales** and add data from the MS Excel file named *SalesData_MultiTabs* including all the worksheets in this file.

Locate this file under the **ExcelFiles** folder.

Apply some data transformations as described below and load the data into the app:

Data transformations:

- In the *products* table:
  - Apply a filter to the table to remove rows related with the "*Free Product Kit*" (with *ProductID* = 90000)
  - **Split** the *Location* field into 3 fields using the dot as a separator: **Floor.Section.Row**
- In the *customers* table:
  - **Rename** the *Contact* field to *Customer_contact*
  - **Replace** the *United Kingdom* values in the *Country* field with *UK*
- In the *orders* table:
  - **Format** the *OrderDate* field using the *MM/DD/YYYY* format
  - **Create a new field** called *Sales* using the expression:

    ```
    Quantity * UnitPrice * (1- Discount)
    ```

  - Bucket the sales into 3 groups creating a new field called *Sales Type*:
    - Low sales - sales lower than the average value of 1180
    - Medium sales - sales greater than or equal to 1180 and lower than 6000
    - High sales - sales greater than or equal to 6000
- In the *employees* table, create *EmpName* as a calculated field using the expression:

  ```
  [First Name] &' '& [Last Name]
  ```

Associate the tables using the recommended associations as follows:

Table associations

- *orders* and *products* by *ProductID*
- *orders* and *customers* by *CustomerID*
- *orders* and *employees* by *EmployeeID* and *EmployeeKey*
- *customers* and *Regions* by *Division*

Load the data and review the final data model and table structure in the Data model viewer.

Hints

- Files can be loaded into an app using the **Files and other sources** option or just by dragging and dropping the file into the app.
- Prepare the data by filtering the rows, renaming fields and tables, and creating new calculated fields from the **Prepare > Data manager > Tables** section. You can also associate the tables identifying the key fields in each with the visual data preparation view (**Data manager > Associations**).
- The **Load data** button loads the tables and fields you define into the app.
- Once the data is loaded in the app, you can review the tables and fields using the **Data model viewer**.
- The ▶**Associations** view in **Data manager** allows you to associate the tables, using the visual data preparation. There are different methods to associate tables:
  - The Ñ button shows the recommended associations and allows you to associate all the tables according to Data manager recommendations. You can break undesired associations afterwards by pulling the bubbles apart until the association breaks.
  - Tables can be associated one by one by dragging them on top of one another. If there is a recommended field/s for the association, the tables get associated using the key field. If there are no recommended associations, you choose the fields from each table and create a custom association.
  - Existing associations between tables can be changed by selecting the joint point between tables. You can change the recommended field or create a custom association.

Step-by-step guide

1. From the Hub, create a new analytics app called *ABC Sales*.
2. Open the new app and add all the data from the *SalesData_MultiTabs* Excel file by dragging and dropping the file into **Files and other sources** area.

   If the data file was loaded automatically, click **Cancel** to continue with Data manager.

   If the **Select data** dialog appears, make sure all tables are selected and the data profiling is Enabled. Then click **Next**.

   > Note: You can add **Filters** when loading the data using the **Filters** tab in the **Select data** dialog or when editing the table in **Data manager**.

3. In the **Data manager**, select the **Tables** ' view and verify that all the tables are displayed in **Pending add** status.

Creating new fields and filtering product data

4. Select the *products* table and click **Edit @**.
5. Select the header in the *Location* column. The values in this field correspond with a 3 number code identifying the *Floor*, *Section*, and *Row* where the product is stored in the shop. In the **Split** section, make sure both dots are selected and click **Create fields**.
6. In the table, rename the new columns as *Floor*, *Section*, and *Row* respectively.
7. Select the first column in the table that represents the product ID. Notice there are a total of 79 identifiers for products. Click **Select data from source** on the top right corner of the dialog. Then click **Filters** (at the top right corner) and select **Add filter** specifying the following filter: *products.ProductID > 90000*.
8. Click **Update**, notice how the total number is now 78 and close the *product* table.

Transforming customer data

9. Select the **@** edit icon in the *customers* table. Then rename the *Contact* column as *Customer_contact*.
10. Notice how the *City* and *Country* fields in this table have been identified as geo fields. Select the *Country* column and review the **Summary** bar chart at the bottom. How many distinct countries appear in the data?
11. In the **Replace** section, select *United Kingdom* and type *UK* as the **Replacement value**. Click **Replace**. How many distinct countries appear now in the data? Close the *customer* table edit dialog.

Preparing order data

12. Edit the *orders* table and review the *OrderDate* field. This field has been identified as a date field. From the Date icon ( ), set the date format for this field from the **Display format** drop-down (*MM/DD/YYYY*).
13. In the *orders* table, click **Add field > Calculated field** to add a new field, *Sales*, using the following expression:

```
Quantity * UnitPrice * (1- Discount)
```

> **Note**: Make sure your expression is the same as above. In Qlik Sense field names are case sensitive.

14. Select the new calculated field and review the metric below the table. Qlik has identified the *Sales* field as a recommended measure and displays some statistics based on the data. Use the transformation cards to classify the sales values into 3 groups: High, Medium, and Low.
15. In the **Bucket** section, at the bottom, create 3 buckets. The default buckets are split equally across the values.
    a. Select the first bucket and set the **Bucket name** as *Low sales* and the **Up to** value to 1180.
    b. Click the right arrow to move to the second bucket and repeat the process using *Medium sales* for the bucket name and 6000 for the upper limit.

c. Click the arrow again to edit the third bucket and set the name to *High sales*. Then select the left arrow and click **Create buckets**.

A new field is created in the table, rename the column as *Sales type*. Then close the *orders* table.

Adding new employee data

16. Edit the *employees* table and add a new calculated field, *EmpName*, using the following expression:

```
[First Name] &' '& [Last Name]
```

Please note the field names are specified using square brackets in this case as these names contain spaces.

> **Note**: You can concatenate strings or field values using the ampersand (&) character. Plain text or strings can also be added using single quotation marks.

17. Review the metrics below the definition to see more details about the new field created. Notice the difference between the count of this field and the distinct values. This difference is due to some duplication of rows in the table (example: review in the table the rows for *EmployeeID E0003*). Then close the *employee* table.

Associating the data

18. Navigate to the **Associations** view to associate all the tables.
19. Associate the tables need as follows:
    - *orders* and *products* are associated using the *ProductID* field in each table.
    - *orders* and *customers* are associated using the *CustomerID* field in each table.
    - *orders* and *employees* are associated using the *EmployeeID* and *EmployeeKey* fields from each table.
    - *customers* and *Regions* are associated using the *Division* field in each table.

The Ñ button shows the recommended associations in a panel on the right, where you can preview them. Click **Apply all** to apply all the recommended associations and load the tables. Navigate to the **Data model viewer** and make sure the final table structure matches the solution diagram.

20. Return to the **Data manager** (**Associations** view) and select the association link between employees and orders. Then rename the association name to *EmployeeID*.
21. **Load data** again to update all the changes and close the dialog after the data is reloaded.

Loading and reviewing the data

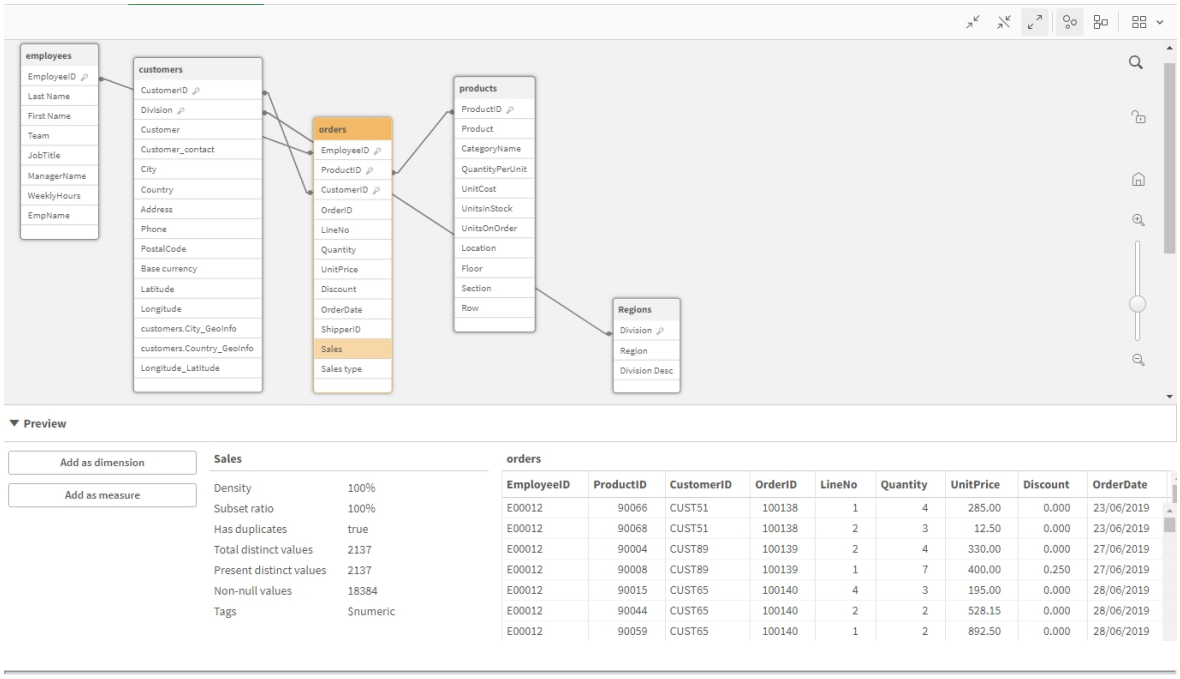
22. Click **Prepare > Data model viewer**. Move the tables so they are visible and do not overlap others. Notice how all the tables are now in the diagram connected to each other.

> **Note**: Use the options in the toolbar to expand all the tables and see the fields.

> **Note**: After loading the *customers* table, new geo fields are created automatically in the table (see fields with *_GeoInfo* suffix). These fields contain the coordinates and area map for the city and country fields. You can review the data in these fields using the **Preview** panel.

23. Select the *orders* table and the *Sales* field in this table and review the **Preview** panel at the bottom to see a sample of data in the *orders* table.

Solution



Activity 2: Define master dimensions and measures

Summary:

- Define new dimensions and measures in apps creating them as master items.
- Understand master dimensions and drill-down dimensions
- Familiarize with the expression syntax and the drill down dimensions.

Qlik Sense app: ABC Sales

Dependencies: Before working in this activity, make sure you complete Activity 1.

Challenge

In the **ABC Sales** app, define reusable master items by creating the following master dimensions and measures.

Master dimensions

Use the specified names for the master dimensions:

Table	Field	Master dimension name
products	*CategoryName*	**Product category**
customers	*Country*	**Country**
customers	*City*	**City**
customers	*Customer*	**Customer**
customers	*Division*	**Division**
employees	*EmpName*	**Sales person**
orders	*OrderDate.autoCalendar.Year**	**Year**
orders	*OrderDate.autoCalendar.Quarter**	**Quarter**
orders	*OrderDate.autoCalendar.Month**	**Month**
orders	*OrderDate.autoCalendar.YearMonth**	**Year Month**
products	*Product*	**Product**

Create two additional master dimensions using the Drill-Down capability:

- **Product Drill-down**: *Region* > *Division* > *CategoryName* > *Product*
- **Year Drill-down**: *OrderDate.autoCalendar.Year** > *OrderDate.autoCalendar.YearMonth**

*If the OrderDate.autoCalendar.*Field* is not available, use the OrderDate.*Field*.

Master measures

Use the following measure names, calculations, and color codes (when specified) for the master measures:

Master measure name	Expression	Color
Sales	`Sum(Sales)`	#58ac11
COGS	`Sum(UnitCost * Quantity)`	#4477aa
Discount	`Num(Avg(Discount),'0.00%')`	# ffcf02
Margin	`Sum(Sales - (UnitCost * Quantity))`	#7db8da
Margin%	`Num(Sum(Sales - (UnitCost * Quantity)) / Sum(Sales),'0.00%')`	#7db8da
Num Orders	`Count(Distinct OrderID)`	#8e477d

Note the **Num()** function is used in some expressions to format the numbers. You can also format some of these expressions using the **Number formatting** option.

Drag the new measures into the sheet to check the expressions are evaluating correctly.

Hints

- Master items can be added to the app in different ways:
  - From the **Assets** panel (visible when editing a sheet), under the **Master items** section
  - From the **Assets** panel (visible when editing a sheet), under the **Fields** section, by right-clicking on the field
  - From the **Data model viewer**, under the **Preview** panel, when the field is selected
  - From the **Properties panel** of an object, when you have selected some field to be used as a measure or dimension, and you want to add it as a master item. Under the **Data** section, when the measure or the dimension is selected, by clicking on **Add new**.
- A hierarchy of fields can be defined as a drill-down master dimension. Drill-down dimensions are created as master dimensions using the **Drill-down** option radio button of the **Create new dimensions** dialog.
- When editing the sheet, you can drag and drop fields and master dimensions and measures into the sheet canvas. The sheet can be analyzed by toggling out of edit mode.

Step-by-step guide

1. In the **ABC Sales** app, navigate to the **Sheet**. The *My new sheet* opens as an empty sheet, because we haven't created any sheets in the app. Click **Edit sheet** to enable the edit mode and edit the sheet.

   Note: You may need to use the **Advanced options** toggle to access the Assets panel in Edit mode.

Defining master dimensions

2. From the **Assets panel**, under **éMaster items**, select **Dimensions** and then click **Create new**.
3. Select one of the fields in the first table above (you can search for the field), and then click **Create**.
4. Repeat steps 2 and 3 for each dimension in the Master dimensions table, changing the dimension name as required.
5. From the **Assets panel**, under **Master items** > **Dimensions**, create a new master dimension.
6. In the **Create new dimensions** dialog, select **Drill-down** and add *Region, Division*, *CategoryName*, and *Product* in that order to insert them into the **Field** list. Name the dimension as *Product drill-down*. Then click **Create**.

> **Note**: You can rearrange the order of the fields you have selected, if necessary, using drag-and-drop in the list of selected fields.

7. Repeat steps 5 and 6 for the *Year drill-down* dimension. Make sure you add the *OrderDate.autoCalendar.Year* and *OrderDate.autoCalendar.YearMonth* in that order.

> **Note:** The *OrderDate.Year* and *OrderDate.YearMonth* fields appear as *OrderDate.autoCalendar.Year* and *OrderDate.autoCalendar.YearMonth* under the **Field** section of the dialog box.

Defining master measures

8. From the **Assets panel**, under **é Master items**, select **Measures**, then click **Create new**.
9. Type the corresponding **Expression** and **Name** for the measure (based on the Master measures table above). You can copy/paste the expression instead of typing it, if needed.
10. If the measure has a specified color, choose the color from the **Measure color** picker.
11. Repeat steps 9 and 10 for each measure in the table. Notice that master measures can be duplicated by right-clicking on them. This can be useful when creating similar measures, like *Margin* and *Margin%*, as you can edit the copied measure.

Reviewing the data

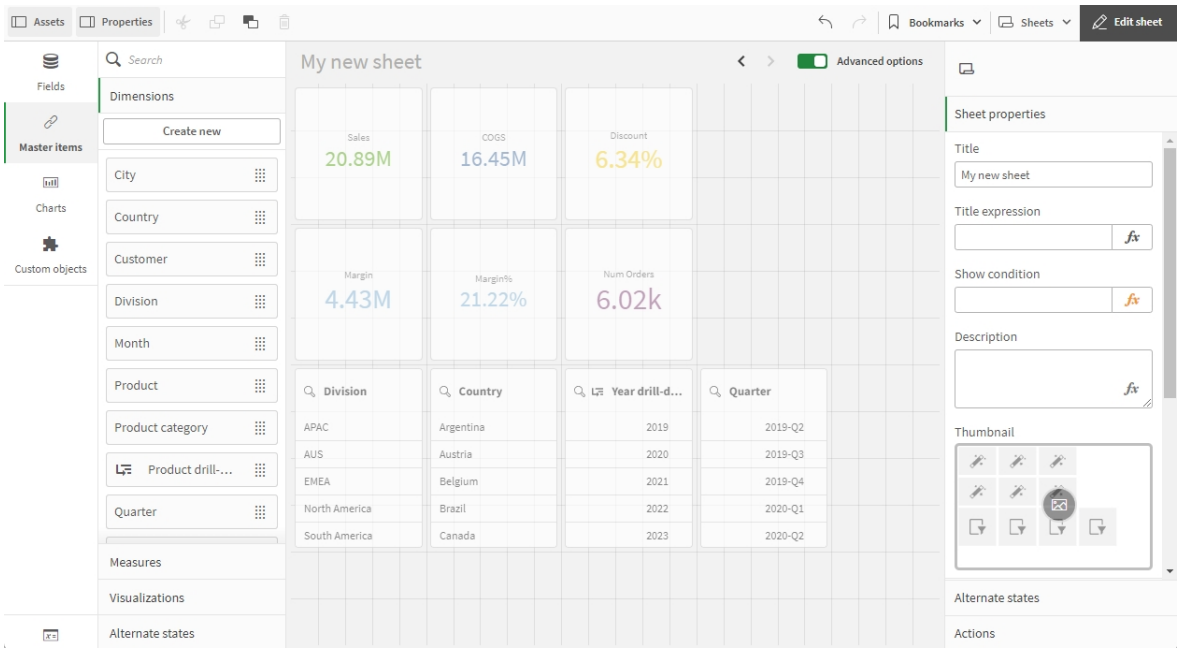
12. Drag and drop the new measures into the sheet and compare the results with the solution image. Notice how the assigned color is displayed for each measure. Make sure you represent all as KPIs. You can use the **Chart suggestions** on the properties panel, by clicking on Auto, and then you will see the **Recommended charts**, where you will be able to select the chart type you prefer.
13. With the **Shift** key pressed, drag and drop some of the dimensions into the sheet to use them as filters.

> **Note**: Add fields or dimensions as filter panes into the sheet by holding the shift key when you drag and drop the items into the sheet. This prevents Chart suggestions. Master expressions defined in the app can be used in chart expressions. You can copy and paste measures between visualizations.

14. Then click **Edit sheet** to enable the analysis mode in the sheet to interact and make selections.

> **Qlik Sense Client-Managed**: Click **Done editing** to come back to the analysis mode.

Solution



Activity 3: Create calendar measures and advanced expressions

Summary:

- Use the Data manager to prepare the tables for analysis by adding source filters, transforming fields, and creating new fields.
- Familiarize with the Visual preparation (Data manager associations) to associate the tables and create the data model.

Qlik Sense app: ABC Sales

Dependencies: Before working in this activity, make sure you complete Activity 1.

Challenge

In the **ABC Sales** app, create the following calendar measures using the specified names:

Calendar measure name	Description
Sales YTD	Sum of *Sales* for the Year to Date (YTD) based on the *OrderDate*.
Sales CYTD	Sum of *Sales* for the Current Year to Date (CYTD) based on the *OrderDate*.
Sales LYTD	Sum of *Sales* for the Last Year to Date (LYTD) based on the *OrderDate*.

Bonus Point:

Include two additional master measures with YTD Flags:

Calendar measure name	Description
Sales CYTD - LYTD	Difference between the Sum of *Sales* for the CYTD and the LYTD.
Num Orders CYTD	Count distinct *OrderID* for the Current Year to Date (CYTD) based on the *OrderDate*.

Hints

- Calendar measures allow you to generate several master measures at once based on time ranges (yearly, quarterly, etc).
- Calendar measures are created from the **Fields** section in the **Assets** panel. Right-click on the field you want to apply the expression, for instance, the *Sales* field, and select **Create calendar measures**.
- You can reuse existing master measures to create a new one by right-clicking the measure and selecting **Duplicate**. Then you can **Edit** the duplicated measure, changing the name and expression.

Step-by-step guide

1. In the **ABC Sales** app, open *My new sheet* in edit mode.

Creating Sales YTD, Sales CYTD, and Sales LYTD

2. In the **Assets** panel, expand the **Fields** list and right-click on the *Sales* field.
3. Select **Create calendar measures**.
4. Select *OrderDate* as **Date field**, *Sum* as the **Aggregation** function and **Yearly** as the **Time range**.
5. Click **Save to master items** and close the window. This option generates, by default, three master measures.
6. In the **Assets panel**, under **é Master items**, expand **Measures** and review the three new measures:
   - **SUM Sales YTD** - for the year to date time range based on today's date (applicable to all the years in the app)
   - **SUM Sales YTD Current Year** - for the year to date based on today's date (and for the current year based on today's year)
   - **SUM Sales YTD Last Year** - for the year to date based on today's date (and for the previous year based on today's year)
7. Edit these master measures to change the **Name** in each of them to *Sales YTD*, *Sales CYTD*, *Sales LYTD* as in the table above.

Bonus Point:

Creating Sales CYTD - LYTD

8. Right-click the **Sales CYTD** field and select **Duplicate**. A copy of the **Sales CYTD** master item is created.
9. Edit the **Sales CYTD (1)** and rename it to *Sales CYTD - LYTD*.
10. Change the **Expression** as follows and click **Save**:

```
[Sales CYTD]-[Sales LYTD]
```

> **Note**: Existing master measures can also be used as part of other expressions by referencing them by their measure labels.

Creating Num Orders CYTD

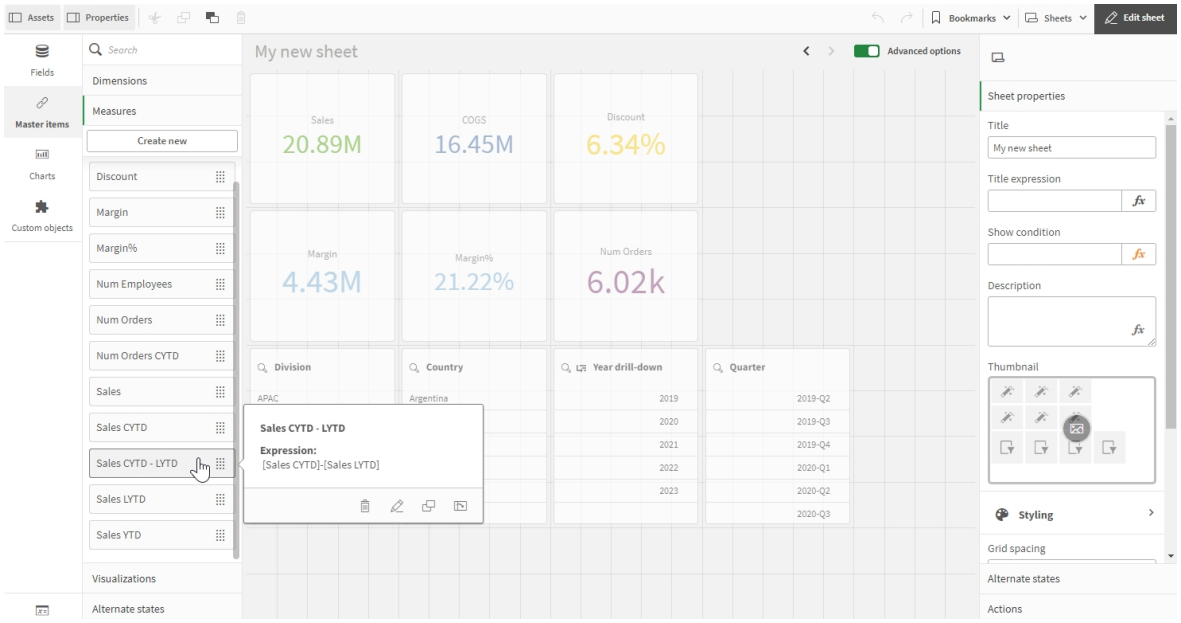
11. Right-click the **Sales CYTD** and select **Duplicate**. A copy of the **Sales CYTD** master item is created.
12. Edit the **Sales CYTD (1)** and rename it to *Num Orders CYTD*.
13. Change the following in the **Expression** and click **Save**:
    a. Change the initial part of the expression "**Sum(**" with "**Count(distinct**"
    b. Change the final part of the expression "**[Sales])**" with "**OrderID)**"

    The final expression looks as follows:

```
Count( Distinct { $ < [OrderDate.autoCalendar.InYTD]={1}, [Order-
Date.autoCalendar.YearsAgo]={0} > } OrderID )
```

Note: The results of the YTD calculations depend on the current date (today).

Solution



Activity 4: Define the business logic

Summary:

- Define the business logic and vocabulary for the app based on the master items.

Qlik Sense app: ABC Sales

Dependencies: Before working in this activity, make sure you complete Activity 1 and Activity 2.

Challenge

In the **ABC Sales** app, create the following reusable items:

Customize the logical model and vocabulary in the app to improve Insight Advisor queries:

- Group all the Sales master measures together in a group of the logical model and define a hierarchy between the *Regions* and *products*.
- Refer to COGS as *Cost* or *Expense* when performing natural language searches.
- Use the term *Iberia* to refer to the following countries together: *Spain* and *Portugal*.
- Refer to Ranking analysis as *Top*, *Best* or *Highest* when performing a search on Insight Advisor.
- Set a predefined question suggested to users when searching for insights in the app data.

Hints

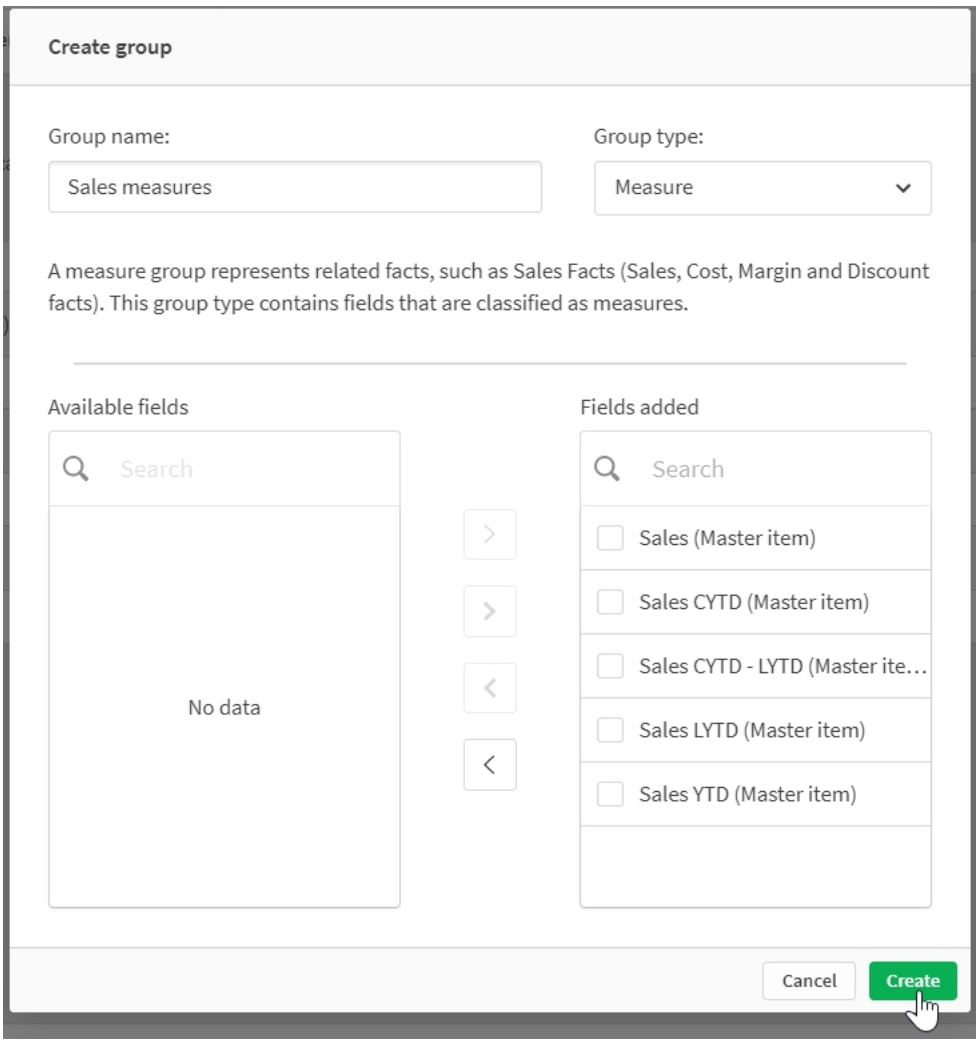
- The logical model and the vocabulary can be customized from the **Prepare > Business logic** sections.
- You can toggle the logical model on and off by clicking **Disable logic/Enable logic**.
- When at the **Analyze < Sheet** section of the app, the **Insight Advisor search** is always available at the top of the app window. You don't need to click on the **Insight Advisor** button first, even if you are editing the sheet, you will be able to use the search bar on the top.

Step-by-step guide

1. In the **ABC Sales** app, navigate to the **Prepare > Business Logic** section.

Logical model

2. Select **Logical model** and click **Continue**. An overview of the default logical model is presented.
3. Navigate to **Fields & groups** and locate the *Discount* field by filtering in the **Name** column.
4. Make sure you set the **avg** operation as **Default aggregation** for this field. In this table, you can also change the visibility, classification or field type, and define if the Insight Advisor can perform a look up in the field values.
5. Find the master measures related with *Sales*, select them and click **Ungroup**.
6. Create a new group by selecting the **Create group** button. Type *Sales measures* as Group name and define the group type as *Measure*, then add all of the Sales measures available using the arrows and click **Create**.

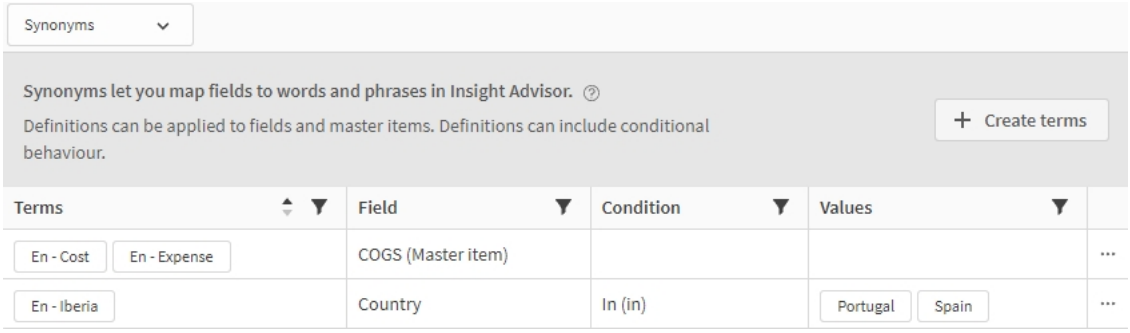


7. Navigate to **Hierarchies** and define a *Region-Products* hierarchy by creating a new hierarchy with that name and adding the *Regions* group at the top of the list, and then adding the *products* group below.

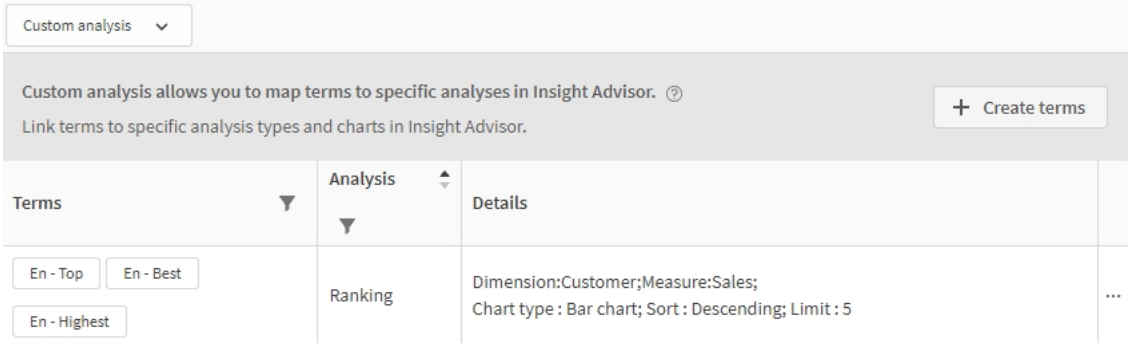
Vocabulary

8. Go to **Prepare > Vocabulary** and navigate to **Synonyms**. Select **Create terms**. Enter the following terms: *Cost* and *Expense* and select *COGS (Master item)* in the **Field** drop-down, then click **Create**.
9. Click **Create terms** again to enter *Iberia*. Select *Country* as **Field** and, for the **Condition**, select *In* and define the following values: *Portugal*, *Spain*.

10. Click **Apply**. Review the vocabulary table with the new two entries.

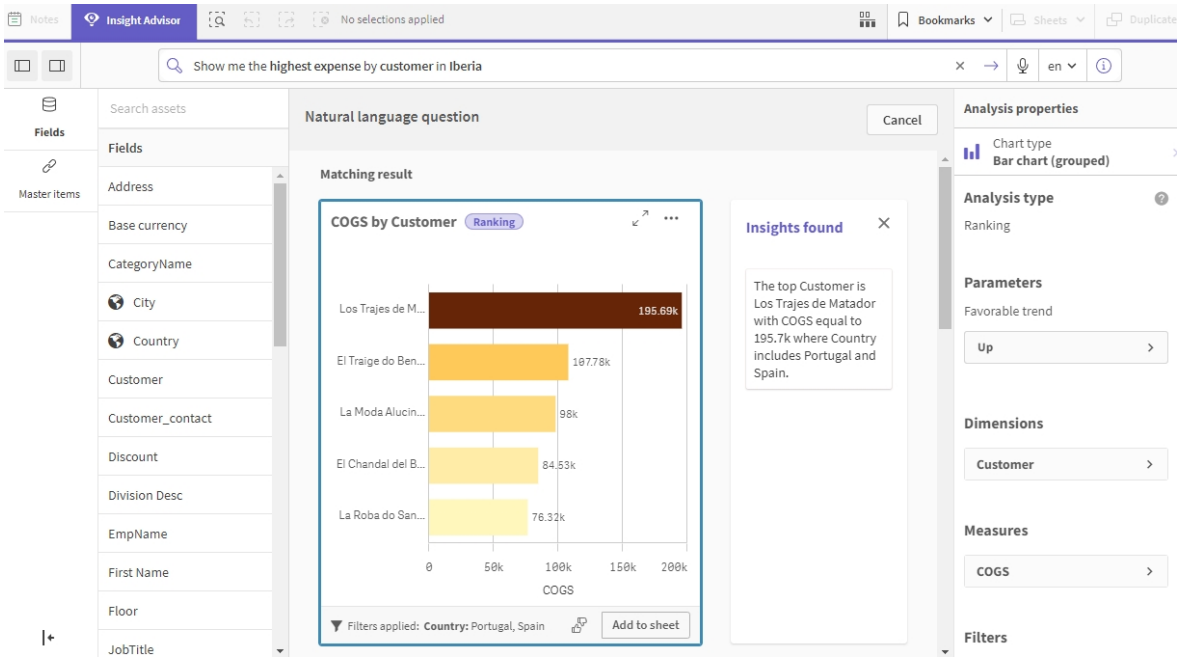


11. Go to **Prepare > Vocabulary** and navigate to **Custom analysis**. Select **Create terms**. Define the following values: *Top*, *Best*, *Highest*. Select **Ranking** from the **Analysis** dropdown.
12. Enter *Customer* as a dimension and *Sales* as a measure on the **Data** tab.
13. Go to **Chart** tab and select Bar chart as **Chart Type**, **sorting** descending and **limit**: 5. Review that everything is ok:



14. Go to **Prepare > Vocabulary** and navigate to **Example questions**. Select **+ Add question**. Confirm the **Language** is set to **En**, and type *Show me the highest expense by customer in Iberia*.
15. Click **Add** and confirm the question is saved.
16. Select **Analyze > Sheet** and navigate to the **Insight Advisor**. Place the cursor in the search bar and see that an example question is suggested by Qlik. Select that question and check the **Matching result**. A chart is built to show *COGS by Customer*, with a filter applied to *Country* and a limitation of *5* customers. This is possible thanks to all the customization we did at the Vocabulary section.

Solution


Notes
Insight Advisor
No selections applied
Bookmarks
Sheets
Duplicate
Show me the highest expense by customer in Iberia
en
Fields
Master items
Search assets
Fields
Address
Base currency
CategoryName
City
Country
Customer
Customer_contact
Discount
Division Desc
EmpName
First Name
Floor
JobTitle
Natural language question
Cancel
Matching result
COGS by Customer
Ranking
Los Trajes de M...
195.69k
El Traige do Ben...
107.78k
La Moda Alucin...
98k
El Chandal del B...
84.53k
La Roba do San...
76.32k
0
50k
100k
150k
200k
COGS
Filters applied: Country: Portugal, Spain
Add to sheet
Insights found
The top Customer is Los Trajes de Matador with COGS equal to 195.7k where Country includes Portugal and Spain.
Analysis properties
Chart type
Bar chart (grouped)
Analysis type
Ranking
Parameters
Favorable trend
Up
Dimensions
Customer
Measures
COGS
Filters